SIMATS ENGINEERING
ASSESSMENT TOOL 3

COURSE NAME: Artificial Intelligence **COURSE CODE: CSA17**

COURSE OUTCOME COVERED

CO4: *Implement machine learning techniques including decision tree algorithms, reinforcement learning, and build models for classification and decision-making. (BL3)*

Assessment Tool Weightage: 50%

Duration : 1 hour

QUESTIONS: 4 **Total Marks:40**

Annexure A

ARTICLE REVIEW

Code of Conduct:

I SR Kaarthika(192412294) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: SR Kaarthika

Annexure A

ARTICLE REVIEW

INSTRUCTIONS

Select any ONE peer-reviewed research article (published after 2019) from IEEE Xplore, Springer, Elsevier, or ACM Digital Library related to any of the following topics: *Inductive Learning, Decision Trees, Statistical Learning Methods (Naïve Bayes / Logistic Regression / SVM), or Reinforcement Learning applications.*

Submit: (i) Article citation details (ii) PDF or valid DOI link (iii) Written review as per the tasks below.

Task 1: Article Summary

(a) Provide the full citation of the article (Author(s), Title, Journal/Conference, Year, DOI). State the domain/application area and the specific ML problem addressed.

(b) Summarise the key objective of the article in your own words (150–200 words). Clearly state the research gap the authors identified and how they proposed to fill it.

Task 2: Methodology Analysis

(a) Explain the ML technique(s) used in the article (e.g., Decision Tree variant, RL algorithm, Statistical model). Describe the dataset used—size, features, source, and how it was prepared.

(b) Map the technique to CO 4 topics (Inductive Learning / Decision Trees / Statistical Learning / RL). Identify one methodological strength and one limitation in the authors' approach.

Task 3: Results & Critical Evaluation

(a) Report the key results (accuracy, F1-Score, AUC-ROC, reward convergence, etc.) as presented by the authors. Create a comparative table if multiple models are benchmarked.

(b) Critically evaluate the results: Are the reported metrics realistic? Are there potential biases (dataset, evaluation protocol)? What additional experiments would strengthen the findings?

(c) Discuss real-world applicability: Where can this technique be deployed? What are the challenges in scaling the solution to industry (data availability, compute, ethical issues)?

Task 4: Reflection & Learning Outcome

(a) State three key insights you gained from reading the article that deepen your understanding of CO 4 topics. Connect each insight to a specific concept from the course syllabus.

(b) If you were to extend this research, propose one novel experiment or enhancement (different dataset / improved algorithm / new application domain). Justify its feasibility and expected impact.

RUBRICS FOR CALCULATION:

Criteria	**Weightage (%)**	**Level 4 (Exemplary)**	**Level 3 (Proficient)**	**Level 2 (Developing)**	**Level 1 (Beginning)**
Understanding of Article	20%	Demonstrates clear and in-depth understanding of the article's purpose, concepts, and arguments	Shows good understanding with minor gaps	Partial understanding; misses key ideas	Lacks understanding of the article
Summary	20%	Provides concise and accurate summary highlighting all key points	Covers most key points with minor omissions	Incomplete or partially accurate summary	Poor or incorrect summary
Critical Analysis	25%	Provides insightful critique with strong reasoning and evaluation	Provides reasonable analysis with some justification	Superficial analysis; limited evaluation	No meaningful analysis
Use of Evidence	15%	Effectively uses examples, data, or references to support review	Uses some supporting evidence with minor gaps	Limited or weak supporting evidence	No supporting evidence provided
Organization	20%	Well-structured, clear, and coherent writing with proper language and flow	Generally clear with minor issues	Poor organization; lacks clarity	Disorganized and difficult to understand

RESEARCH ARTICLE REVIEW

Selected Article

Adua, E., Kolog, E. A., Afrifa-Yamoah, E., Amankwah, B., Obirikorang, C., Anto, E. O., Acheampong, E., Wang, W., & Tetteh, A. Y. (2021). "Predictive model and feature importance for early detection of type II diabetes mellitus." Translational Medicine Communications, 6, Article 17. DOI: 10.1186/s41231-021-00096-z.

The article was published on 23 August 2021 in *Translational Medicine Communications* by Springer Nature and is available as an open-access article.

DOI: 10.1186/s41231-021-00096-z

Domain: Healthcare / Medical Diagnosis

ML Problem: Early prediction and identification of Type II Diabetes Mellitus (T2DM) using machine learning classification algorithms.

Task 1: Article Summary

(a) Citation, Domain and ML Problem

The study focuses on using machine learning to predict Type II Diabetes Mellitus in a Ghanaian population. The authors compared four classification techniques: Naïve Bayes, K-Nearest Neighbour, Support Vector Machine and Decision Tree. They also investigated which patient characteristics were most useful for predicting diabetes.

The main application area is healthcare, specifically early diabetes detection and risk prediction.

(b) Key Objective and Research Gap

The main objective of the research was to determine whether machine learning algorithms could accurately predict Type II Diabetes Mellitus and identify the most important clinical and anthropometric factors associated with the disease. The authors focused on a Ghanaian population because diabetes prediction models developed in one population may not necessarily generalise well to another population. They compared four machine learning classifiers—Naïve Bayes, KNN, SVM and Decision Tree—and ranked the predictors using Recursive Feature Elimination. The research gap identified was that conventional approaches such as logistic regression may not adequately capture complex relationships among multiple diabetes risk factors. The authors therefore investigated whether machine learning could provide stronger predictive performance while also identifying important biomarkers. The study found that HbA1c, total cholesterol and BMI were the three highest-ranked predictors. Naïve Bayes achieved the best overall performance, with 87% accuracy and an AUC of approximately 0.87. The authors concluded that machine learning could support earlier detection and more targeted diabetes surveillance, particularly in settings where medical expertise and resources are limited.

Task 2: Methodology Analysis

(a) ML Techniques and Dataset

The authors used four supervised classification algorithms:

1. Naïve Bayes (NB)

2. K-Nearest Neighbour (KNN)
3. Support Vector Machine (SVM)
4. Decision Tree (DT)

Dataset

The dataset contained:

- 438 participants
- 219 T2DM patients
- 219 healthy controls
- 11 attributes
- 10 predictor variables
- 1 target variable

The predictor variables were:

- Age
- BMI
- Systolic Blood Pressure (SBP)
- Diastolic Blood Pressure (DBP)
- HbA1c
- Fasting Blood Sugar (FBS)
- Total Cholesterol (TC)
- Triglycerides (TG)
- HDL-C
- LDL-C

The target variable was T2DM status.

The diabetic participants were recruited from Komfo Anokye Teaching Hospital, while healthy controls were recruited from communities in the Kumasi metropolitan area.

Data Preparation

The authors:

- Checked for outliers using boxplots.
- Found 17 missing observations.
- Used the Expectation-Maximization (EM) algorithm to handle missing values.
- Scaled the predictor variables to the range 0–1.
- Used Recursive Feature Elimination (RFE) for feature ranking.
- Used an 80:20 train-test split.
- Maintained equal numbers of diabetic and control samples in the training set.

(b) Connection to CO-4 Topics

CO-4 Topic	Connection to Article
Inductive Learning	Models learn patterns from labelled diabetic and non-diabetic examples.
Decision Trees	DT was used as one of the four classification algorithms.
Statistical Learning	Naïve Bayes is a probabilistic statistical learning method.
SVM	SVM was used for binary classification.
Feature Selection	RFE was used to rank important predictors.

Methodological Strength

A major strength is that the researchers compared multiple ML algorithms on the same dataset rather than relying on only one model. They also performed feature-importance analysis, which makes the results more useful for medical interpretation.

Methodological Limitation

The major limitation is the small sample size of only 438 participants. The authors themselves acknowledged that the small sample could cause the predictions to be overestimated or underestimated.

Task 3: Results & Critical Evaluation

(a) Key Results

The reported overall accuracies were:

Model	Accuracy	AUC-ROC
Naïve Bayes	87%	0.87
SVM	84%	0.84
KNN	83%	0.85
Decision Tree	81%	0.81

Naïve Bayes was the best-performing model according to the authors. It detected approximately 82% of diabetic test cases, with 93% precision for the detected diabetic cases, and its diabetic-class F1-score was 87%.

The authors reported that all four classifiers had AUC values above 0.80, which they interpreted as good discrimination.

(b) Critical Evaluation

The results are promising, but they should not automatically be interpreted as evidence that the models are ready for clinical deployment.

Potential Biases

1. Small dataset:
Only 438 participants were included. A larger dataset would provide more reliable estimates.

2. Sampling bias:
The diabetic participants came from a hospital while healthy controls were recruited from selected communities. This may not represent the general Ghanaian population.

3. Balanced dataset:
The study used equal numbers of diabetic and healthy participants. Real-world populations may have different prevalence rates, so performance could change in practice.

4. Limited external validation:
The models were evaluated using the study dataset rather than being extensively validated on an independent population.

Additional Experiments

The research could be strengthened by:

- Using a much larger dataset.
- Performing external validation using another hospital.
- Using repeated stratified cross-validation.
- Testing the models on populations from different regions.
- Reporting calibration in addition to discrimination.
- Testing sensitivity at clinically relevant thresholds.
- Including additional predictors such as family history, physical activity and diet.

The authors themselves noted that family history and physical activity were not included and could potentially improve the models.

(c) Real-World Applicability

The technique could potentially be deployed in:

- Hospitals
- Diabetes screening centres
- Primary healthcare centres
- Rural healthcare facilities
- Electronic health-record systems
- Clinical decision-support systems
- Public-health screening programmes

The main benefit is that ML could help identify people at higher risk and support earlier clinical assessment.

Industry Challenges

Data availability: Medical data may be incomplete or inconsistent.

Generalisation: A model trained using Ghanaian patients may not perform equally well on Indian, European or other populations.

Privacy: Patient health information must be securely stored and processed.

Explainability: Doctors need to understand why a patient has been classified as high risk.

Clinical validation: The model needs extensive independent testing before clinical use.

Bias: Models can inherit biases from the population represented in their training data.

Task 4: Reflection & Learning Outcome

(a) Three Key Insights

Insight 1 – Inductive Learning

I learned that supervised ML performs inductive learning by learning general patterns from labelled examples and applying those patterns to previously unseen patients.

This connects directly to the CO-4 concept of Inductive Learning.

Insight 2 – Statistical Learning

The study demonstrated how Naïve Bayes can be used for medical classification. Instead of simply producing a decision rule, it estimates the probability of a patient belonging to a particular class.

This connects to Statistical Learning Methods – Naïve Bayes.

Insight 3 – Decision Tree and Feature Importance

I learned that a Decision Tree can classify patients through a sequence of feature-based decisions, while feature-selection techniques can identify which variables contribute most to prediction.

In this study, HbA1c, total cholesterol and BMI were identified as the top three predictors across the models.

This connects to Decision Trees and Feature Selection in CO-4.

(b) Proposed Extension

I would extend this research by developing a larger multi-hospital diabetes prediction model using an Indian healthcare dataset.

The new study could combine:

- HbA1c
- Fasting glucose
- BMI
- Blood pressure
- Cholesterol
- Age
- Family history
- Physical activity
- Diet
- Medication history

I would compare Naïve Bayes, Decision Tree, Logistic Regression, Random Forest and XGBoost using repeated cross-validation and independent external validation.

The proposed experiment is feasible because these clinical variables are commonly collected during diabetes screening. It would also improve the model's ability to generalise to a different population and provide a more realistic evaluation of its potential for clinical deployment.

Conclusion

The selected research demonstrates that machine learning can be useful for early detection of Type II Diabetes Mellitus. Among the four tested algorithms, Naïve Bayes achieved the best overall performance with 87% accuracy and approximately 0.87 AUC, while Decision Tree achieved 81% accuracy and 0.81 AUC. The study also identified HbA1c, total cholesterol and BMI as the three most important predictors. However, the small dataset and population-specific sampling limit the generalisability of the findings. Therefore, larger datasets, external validation and additional clinical variables are required before such models can be safely deployed in real-world healthcare.